

15 Graph Theory and Network

15.4 Laplacian Matrix and Spectral Clustering

Build a Laplacian matrix from the adjacency and degree matrices, normalize it, eigendecompose it, and use the eigenvectors for the smallest eigenvalues to split a social network into clusters



Core Skills You Will Master

1

The basic pipeline of spectral clustering: adjacency matrix, degree matrix, Laplacian matrix, normalized Laplacian matrix, then eigendecomposition

3

The degree matrix of an undirected graph is a diagonal matrix; its diagonal entries are node degrees, reflecting how active each node is

5

The Laplacian matrix is the degree matrix minus the adjacency matrix, and captures the graph's connection structure

7

Eigenvectors for the smallest eigenvalues place nodes that are close in the resulting low-dimensional space when their social ties are tight

2

For an unweighted undirected graph, the adjacency matrix is symmetric, with 1 marking a connection and 0 marking none

4

High-degree nodes are usually the hub figures of a social network; low-degree nodes tend to sit on the periphery

6

Normalization removes the effect of differing node degrees, preventing highly connected nodes from dominating the clustering result

SECTION ROADMAP

From adjacency and degree matrices, to the Laplacian matrix, to spectral clustering

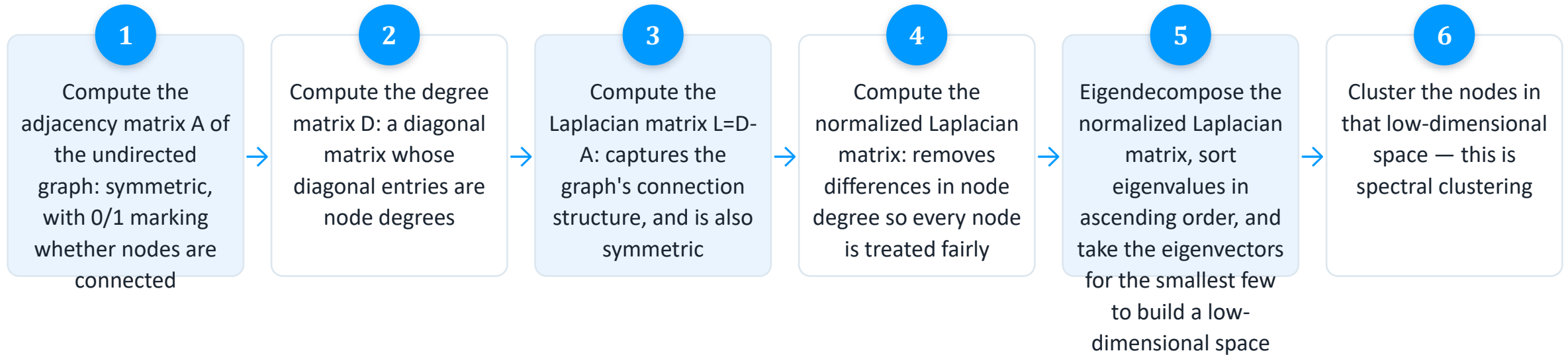


Figure 1 · An Undirected Graph of Karate Club Members

- Nodes are club members, numbered 0–33; edges mark whether two members often trained or socialized together — this is the classic Zachary's Karate Club dataset (proposed in 1977) used to study network structure
- Because of a conflict between the club's lead and a member, the club split that community-detection algorithms on the undirected graph can recover automatically

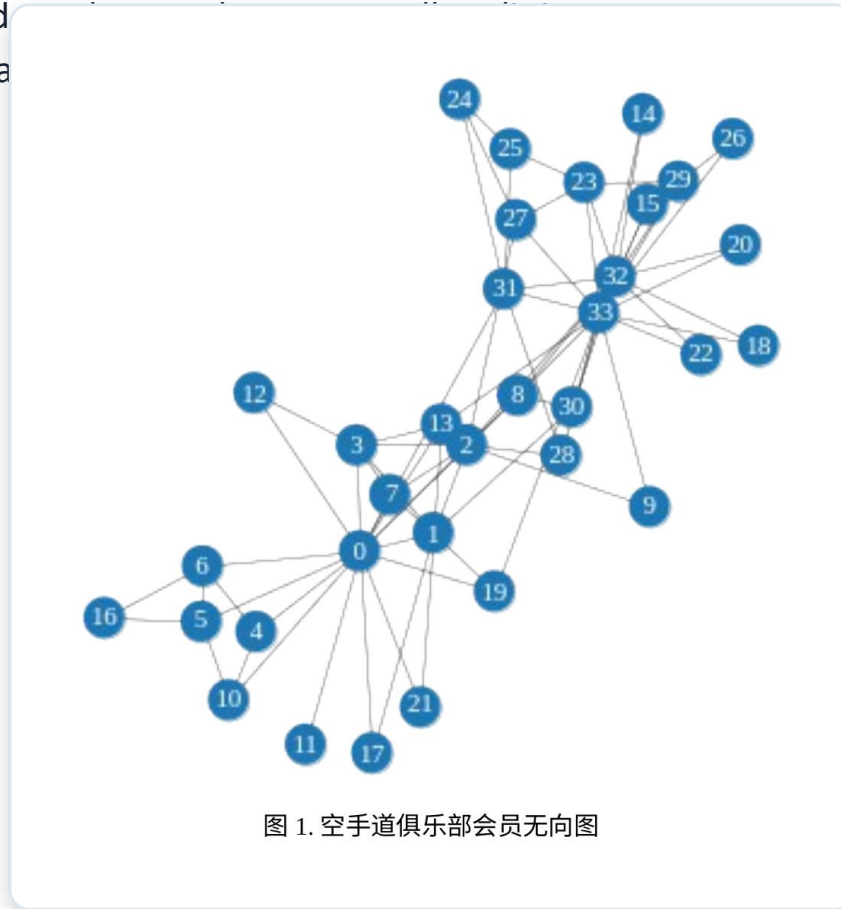


Figure 2 · Adjacency Matrix A

- Adjacency matrix A is computed from the connections in Figure 1; the adjacency matrix of an undirected graph is symmetric
- For an unweighted undirected graph, entry 0 in A means no edge exists between two nodes, and entry 1 means an edge exists

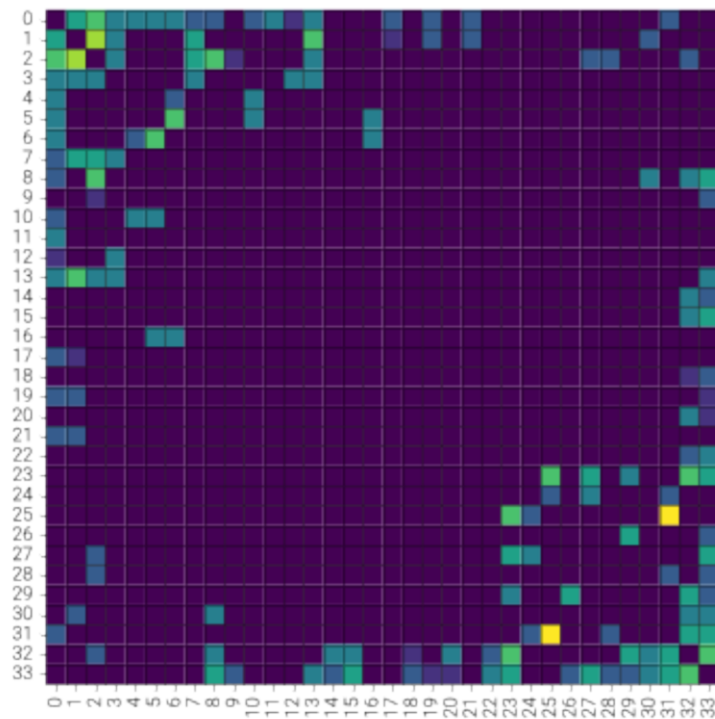


图 2. 邻接矩阵

Figure 3 · Degree Matrix D

- D is a diagonal matrix; each diagonal entry is the degree of the corresponding node — the total number of edges connected to it
- The diagonal entries of D can be obtained directly by summing adjacency matrix A along its rows (or columns)

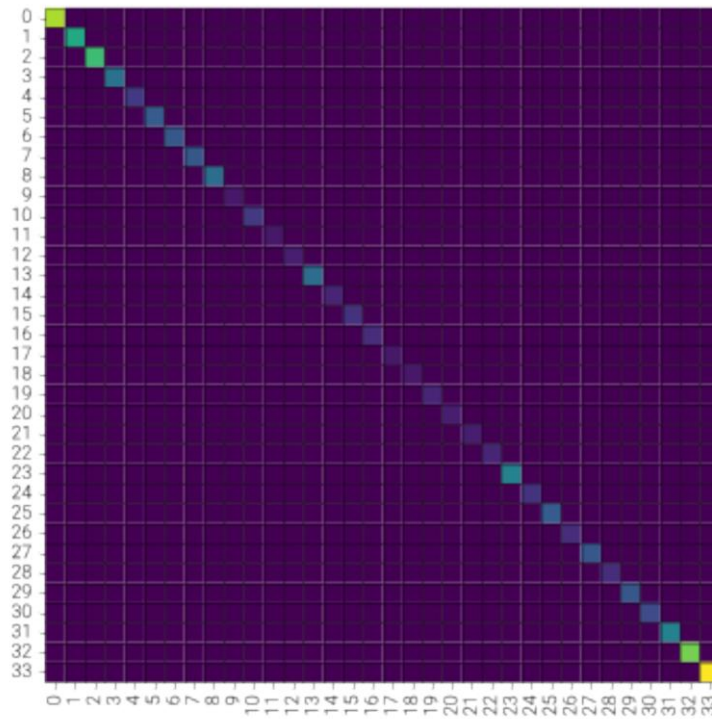


图 3. 度矩阵

Figure 4 · The Degree of Each Node

- Nodes 0 and 33 have especially high degree — they act as the "hub figures" of the group, directly connected to many other members
- A lower degree means a node has fewer ties in the network; it may play a peripheral role or sit in a structurally isolated position

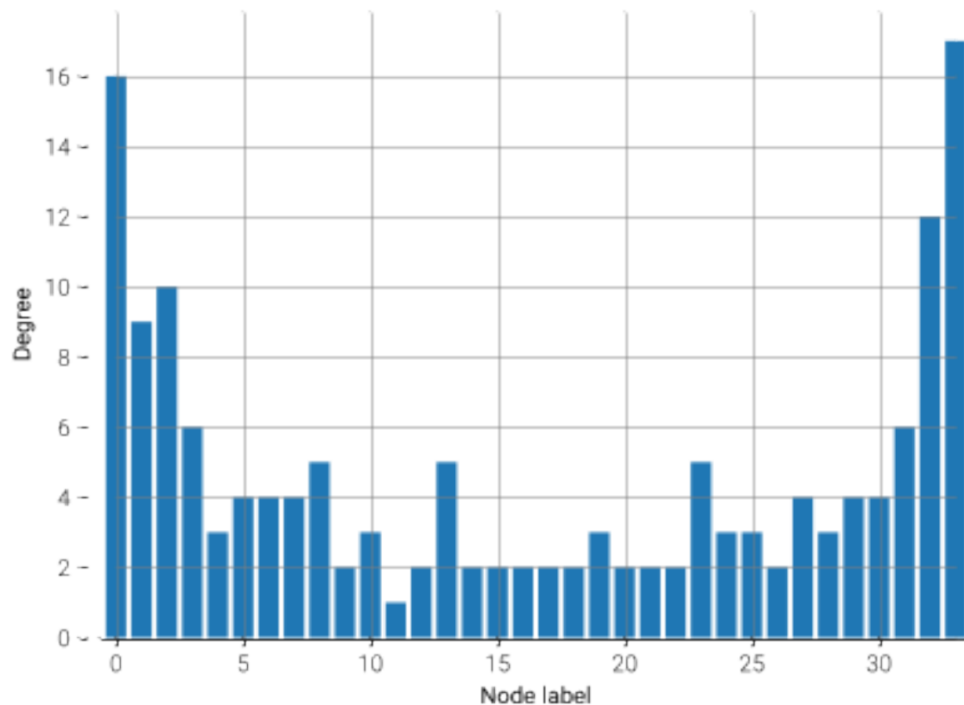
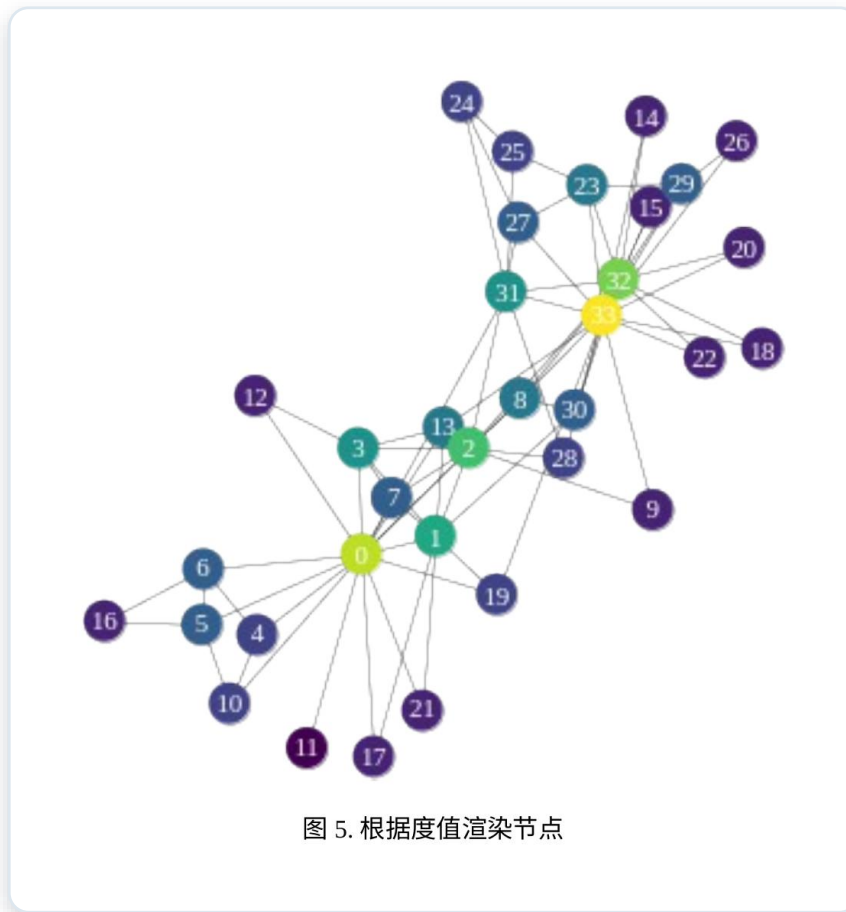


图 4. 各个节点的度

Figure 5 · Nodes Rendered by Degree Value

- Color shade encodes each node's degree; degree reflects a node's direct influence and level of participation in the network
- Looking at the color pattern, two "friend circles" already begin to emerge



The Laplacian Matrix Is the Degree Matrix Minus the Adjacency Matrix

$$L = D - A$$

- For an undirected graph, both degree matrix D and adjacency matrix A are symmetric, so Laplacian matrix L is symmetric too
- The Laplacian matrix is an important representation of graph structure, capturing how nodes are connected; its eigenvalues and eigenvectors can also reveal the network's partition structure

Figure 6 · Laplacian Matrix L

- The diagonal entries of L are each node's degree (from D); the off-diagonal entries are the negative of the corresponding entries of adjacency matrix A
- The Laplacian matrix simultaneously encodes the graph's structure and its connectivity pattern (off-diagonal)

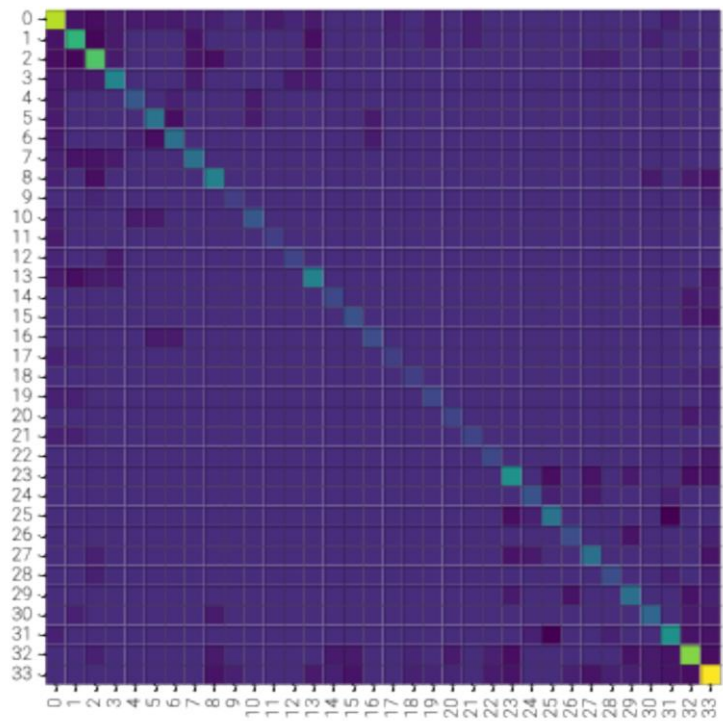


图 6. 拉普拉斯矩阵

Figure 7 · Normalized Laplacian Matrix

- Scaling L on both sides by D to the power of negative one-half — equivalent to rescaling L 's rows and columns — gives the normalized Laplacian matrix, whose diagonal entries are all 1
- In a graph with widely varying node degrees, the high-degree nodes tend to dominate the overall structure; normalization standardizes each node's connection strength

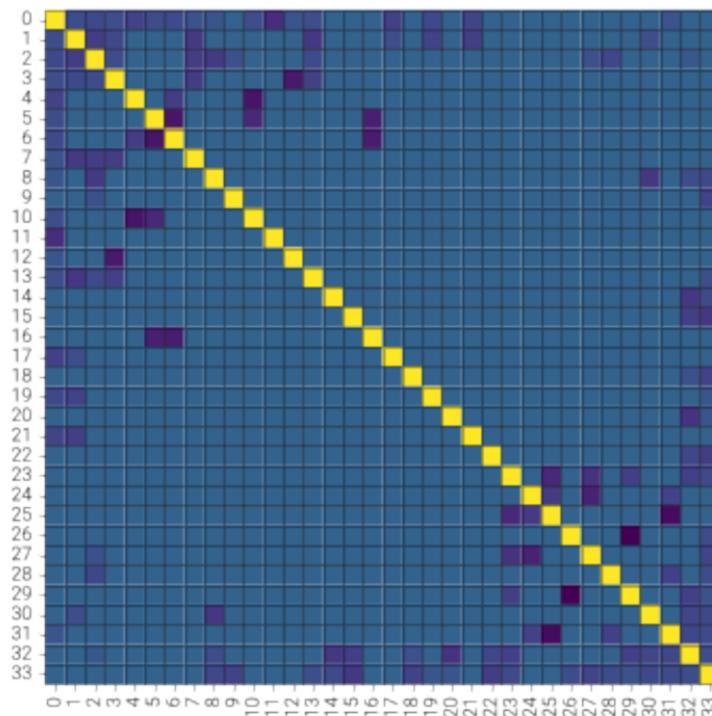


图 7. 归一化拉普拉斯矩阵

Eigendecomposing the Normalized Laplacian Matrix Is a Spectral Decomposition

$$L = V\Lambda V^T$$

- The normalized Laplacian matrix is real and symmetric, so its eigendecomposition is a spectral decomposition: V is an orthogonal matrix, Λ is the diagonal matrix of eigenvalues
- The corresponding eigenvectors are sorted by ascending eigenvalue, in preparation for building a low-dimensional space

Figure 8 · Eigendecomposition of the Normalized Laplacian Matrix (Spectral Decomposition)

- The four matrices in $L=V@Λ@V^T$ are shown in sequence; the columns of V are eigenvectors, and the diagonal of $Λ$ holds the corresponding eigenvalues
- Since the normalized Laplacian matrix is real and symmetric, V is orthogonal, satisfying $V@V^T$ equal to the identity matrix

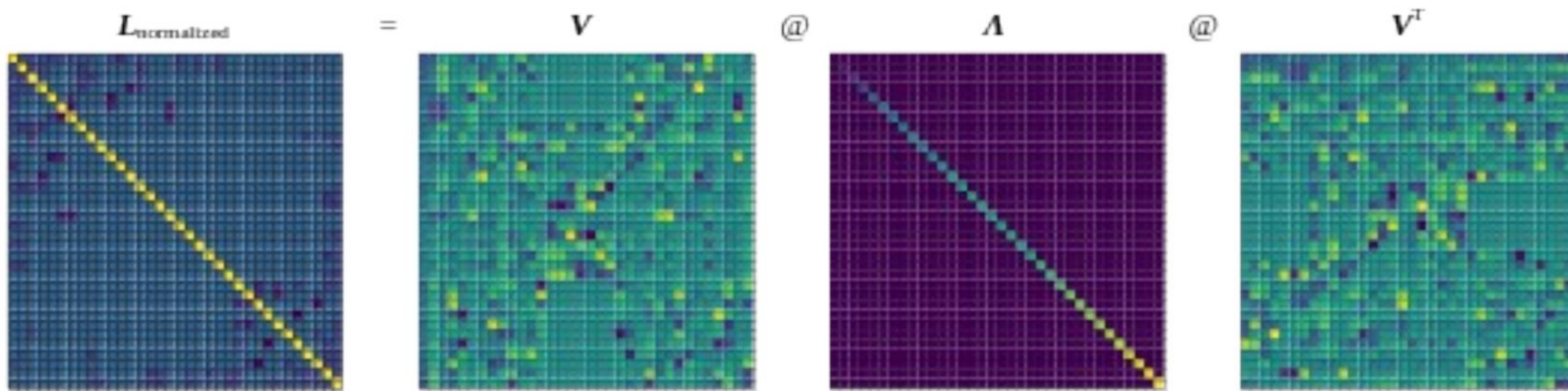


图 8. 归一化拉普拉斯矩阵的特征值分解 (谱分解)

Figure 9 · Diagonal Matrix of Eigenvalues Λ

- The diagonal entries are the 34 eigenvalues, arranged in ascending order from top-left to bottom-right
- All of these eigenvalues are nonnegative — an important property of the (normalized) Laplacian matrix

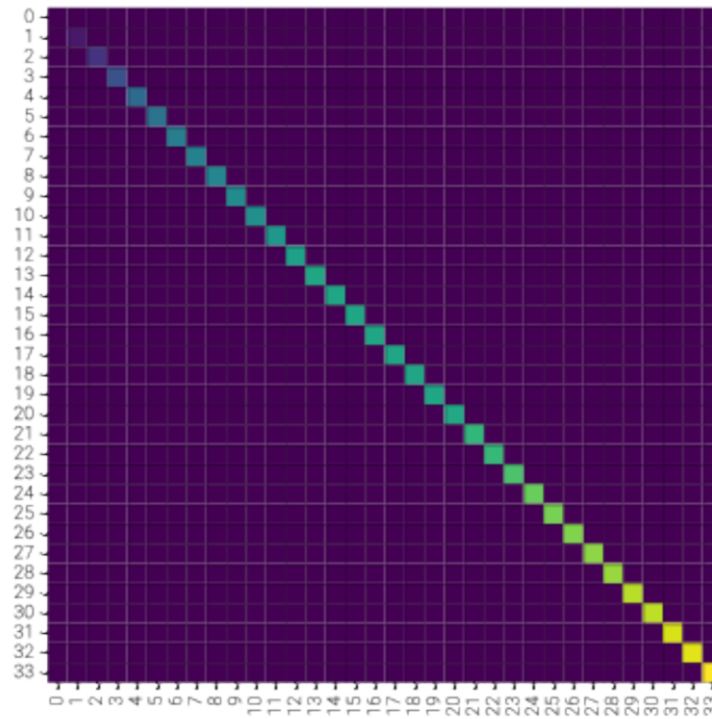


图 9. 特征值对角方阵

Figure 10 · Eigenvalues in Ascending Order

- Sorting the 34 eigenvalues from smallest to largest, the smallest few are clearly below 1, then a cluster sits near 1, before gradually increasing
- Spectral clustering takes the eigenvectors for the smallest few eigenvalues to build a low-dimensional space

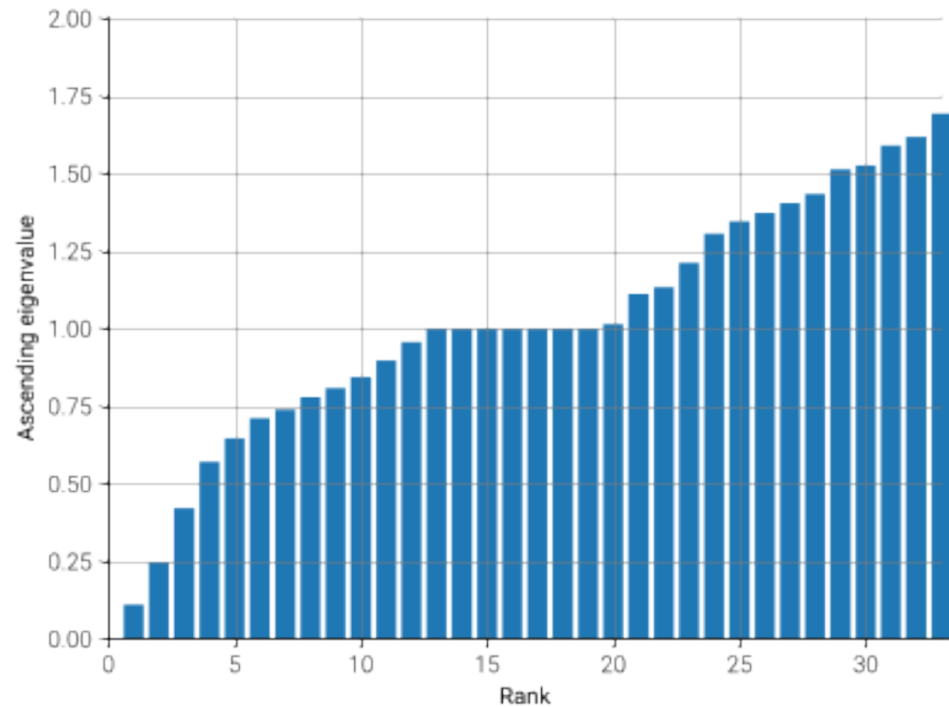


图 10. 特征值递增排列

Figure 11 · Orthogonal Matrix V

- Every column of V is an eigenvector of the normalized Laplacian matrix, and the columns are pairwise orthogonal
- The first two columns, v_1 and v_2 , correspond to the two smallest eigenvalues and will be used to build a 2D representation of each node

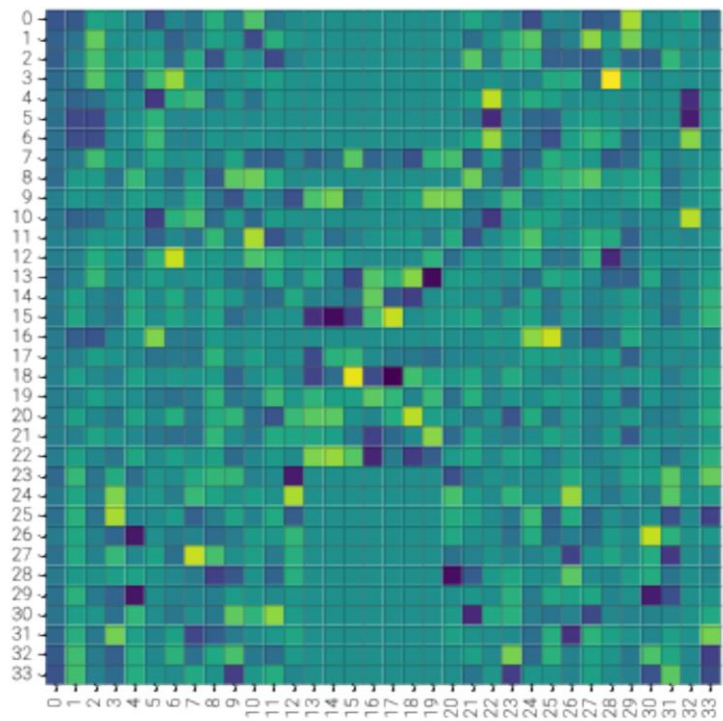


图 11. 正交矩阵 V

Figure 12 · A Scatter Plot Built From the First Two Eigenvectors

- Using the values of eigenvectors v_1 and v_2 as each node's new coordinates: red points have a positive vertical coordinate, blue points negative
- Tightly connected nodes sit close together in this low-dimensional space, while sparsely connected nodes sit farther apart — the 34 members have already been split into two clusters

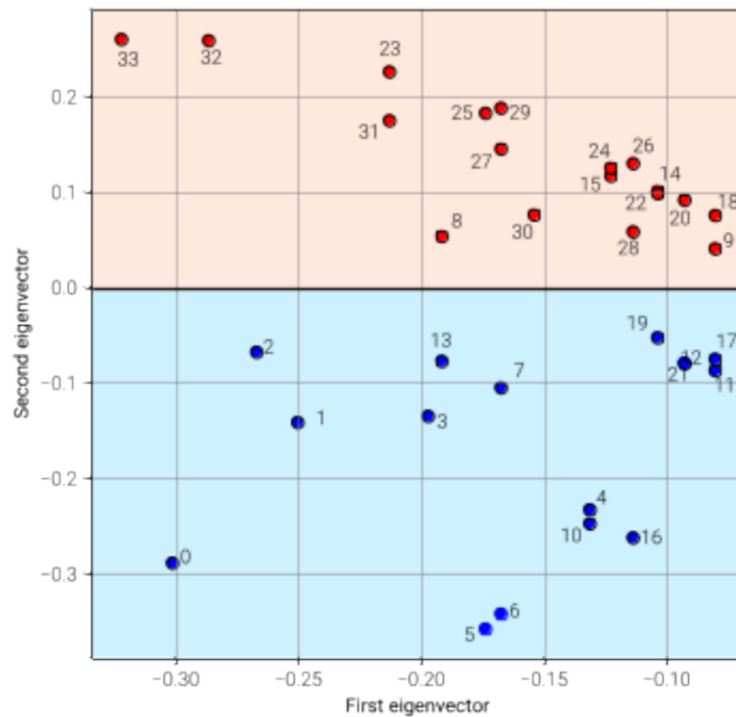
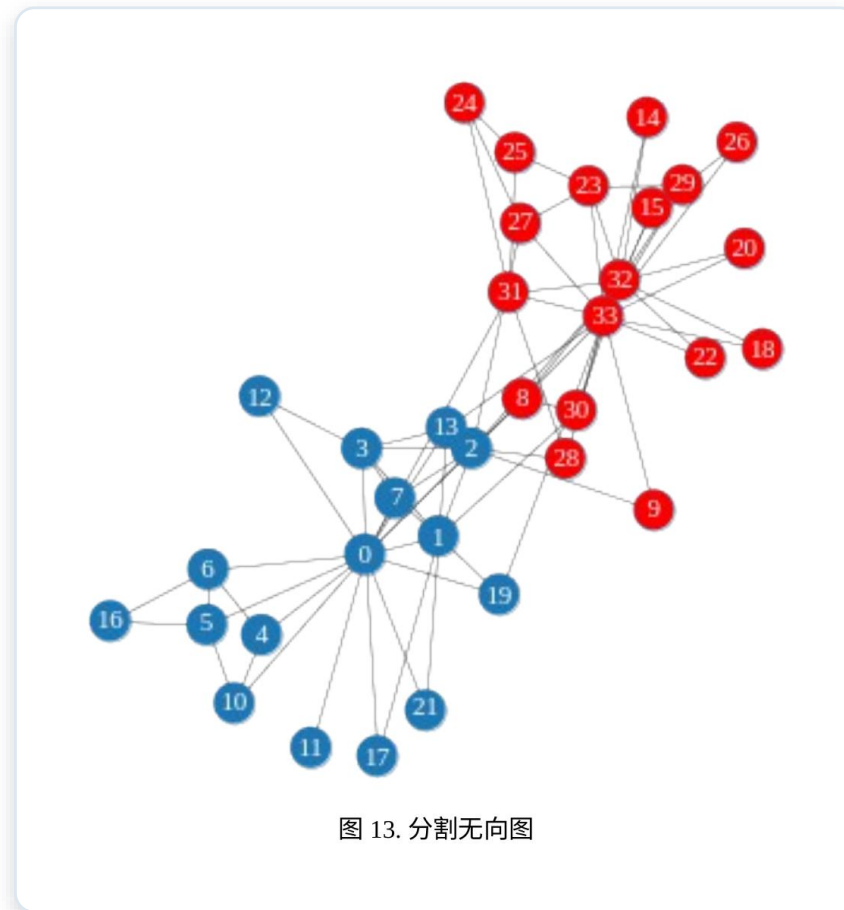


图 12. 前两个特征向量构成的散点图

Figure 13 · The Split Undirected Graph

- Coloring nodes by the sign of their vertical coordinate in Figure 12: red and blue mark the two clusters found by spectral clustering
- This result closely matches the real split Zachary recorded in the club — spectral clustering successfully recovered the social network's split purely from its connection structure



SUMMARY

Build the Laplacian Matrix From the Adjacency and Degree Matrices; Spectral Decomposition Drives Clustering

- ① The adjacency matrix $\mathbf{A}$ of an undirected graph is symmetric; 0 and 1 mark whether an edge exists between two nodes
- ② Degree matrix $\mathbf{D}$ is diagonal; its diagonal entries are node degrees, reflecting how active each node is
- ③ Laplacian matrix $\mathbf{L} = \mathbf{D} - \mathbf{A}$ is also symmetric, and captures the graph's connection structure
- ④ Normalization removes differences in node degree, preventing highly connected nodes from dominating the clustering result
- ⑤ Eigendecomposing (spectrally decomposing) the normalized Laplacian and taking the eigenvectors for the smallest few eigenvalues builds a low-dimensional space
- ⑥ Spectral clustering: adjacency matrix $\rightarrow$ degree matrix $\rightarrow$ Laplacian matrix $\rightarrow$ normalization $\rightarrow$ eigendecomposition $\rightarrow$ low-dimensional space $\rightarrow$ clustering, all in one pipeline

Key Terms in This Section

Adjacency matrix A *The symmetric matrix of an undirected graph; entry 1 marks an edge between two nodes*

Laplacian matrix L *$L = D - A$, capturing the graph's connection structure*

Spectral decomposition *The eigendecomposition of a real symmetric matrix, $L = V\Lambda V^T$*

Degree matrix D *A diagonal matrix whose diagonal entries are node degrees*

Normalized Laplacian matrix *L scaled by D to the power of negative one-half; diagonal entries are all 1*

Spectral clustering *Building a low-dimensional space from the eigenvectors for the Laplacian's smallest eigenvalues, then clustering the nodes*

FURTHER READING

Further Reading

Review · §15.3 Transition Matrix

Compare the transition matrix and the Laplacian matrix, both built from the adjacency and degree matrices

Review · Eigendecomposition

Revisit the spectral decomposition $V\Lambda V^T$ of a real symmetric matrix

Further Reading · Zachary's Karate Club

The classic 1977 social-network dataset — search for more on community-detection algorithms

Open-Source Code Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1 Use NetworkX's built-in `karate_club_graph` to load the karate club dataset, and draw the undirected graph in Figure 1.

Q3 Use Python to compute the degree matrix D from adjacency matrix A , and verify whether D is diagonal.

Q5 Use Python to verify whether every row of Laplacian matrix L sums to 0, and explain why.

Q7 Use Python to eigendecompose the normalized Laplacian matrix, and verify whether all its eigenvalues are nonnegative.

Q9 Use Python to extract the eigenvectors for the 2 smallest nonzero eigenvalues, and plot a scatter plot like Figure 12.

Q2 Use Python to compute the adjacency matrix A of Figure 1, and verify whether A is symmetric.

Q4 Use Python to compute the Laplacian matrix $L=D-A$, and verify whether L is symmetric.

Q6 Use Python to compute the normalized Laplacian matrix, and verify whether its diagonal entries are all 1.

Q8 Use Python to sort the eigenvalues in ascending order and plot a bar chart like Figure 10.

Exercises Q10 – Q17

Q10 Use `sklearn.cluster.SpectralClustering` to cluster the Figure 1 dataset directly, and compare with your manual implementation.

Q12 Use Python to compute the trace of Laplacian matrix L , and explain its relationship to the total node degree.

Q14 Use Python to compare low-dimensional spaces built from the 2 versus the 3 smallest eigenvalues of the Laplacian matrix; do the clustering results match?

Q16 Use Python to count the members in each of the two communities (red and blue) in Figure 1, and compare with Zachary's original split.

Q11 Use Python to verify whether orthogonal matrix V satisfies $V@V^T$ equal to the identity matrix.

Q13 Design your own small undirected graph with 2 clearly separated communities, and repeat this section's full spectral clustering pipeline on it.

Q15 Use Python to verify whether the Laplacian matrix's smallest eigenvalue is always 0, and describe what's special about its eigenvector.

Q17 Use Python to plot the degree distribution histogram for Figure 1, and discuss whether it resembles a scale-free network.

Exercises Q18 – Q25

Q18 Implement k-means clustering directly on the rows of the adjacency matrix using Python, and compare the result with spectral clustering.

Q20 Use Python to verify: if an undirected graph has k connected components, is the multiplicity of eigenvalue 0 of its Laplacian matrix exactly k ?

Q22 Give examples of other real-world networks that could be analyzed with spectral clustering, such as social networks, recommendation systems, or image segmentation.

Q24 Explain why PCA selects the direction for the largest eigenvalue while spectral clustering selects directions for the smallest eigenvalues — what's different about their goals?

Q19 Explain what advantage the normalized Laplacian matrix has over the raw Laplacian matrix for clustering.

Q21 Use Matplotlib to plot heatmaps of adjacency matrix A , Laplacian matrix L , and the normalized Laplacian matrix, and compare the three.

Q23 Use Python to build low-dimensional spaces from the top 2 versus top 3 smallest eigenvectors for the Figure 1 dataset, and compare the visualizations.

Q25 Summarize this section's full pipeline from adjacency matrix to spectral clustering, and note which linear algebra tool each step relies on.

10 Prompts to Explore With AI

1

Explain the basic pipeline of spectral clustering: from adjacency and degree matrices to the Laplacian matrix, normalized Laplacian matrix, and eigendecomposition.

2

Explain how the Laplacian matrix $L=D-A$ is constructed, and why it captures a graph's connection structure.

3

Explain why the Laplacian matrix needs to be normalized, and how clustering results differ before and after normalization.

4

Explain why spectral clustering uses the eigenvectors for the smallest eigenvalues, while PCA uses the direction for the largest eigenvalue.

5

Explain the origin of the Zachary's Karate Club dataset, and why it's often used to test community-detection algorithms.

10 Prompts to Explore With AI

- 6 Explain what special meaning the eigenvector for eigenvalue 0 of a Laplacian matrix has, and how it relates to graph connectivity.
- 7 Give examples of other scenarios where spectral clustering could be applied, such as social networks, image segmentation, or recommendation systems.
- 8 Explain the relationship among the degree matrix, adjacency matrix, and Laplacian matrix, and what property of the graph each one captures.
- 9 Compare spectral clustering with k-means clustering, and explain spectral clustering's advantage when clusters are non-convex.
- 10 Looking back over this entire Linear Algebra book, describe how linear algebra builds up a toolbox for real problems — from vectors and matrices to eigendecomposition and spectral clustering.

Truly Understanding — and Using — Linear Algebra

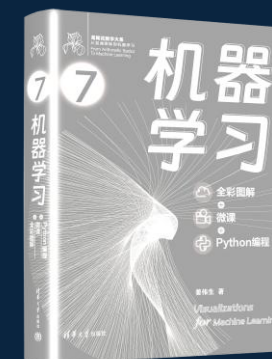
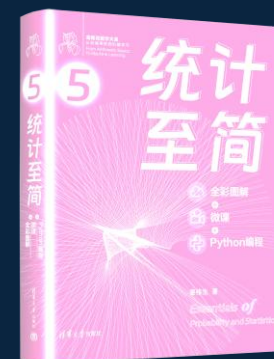
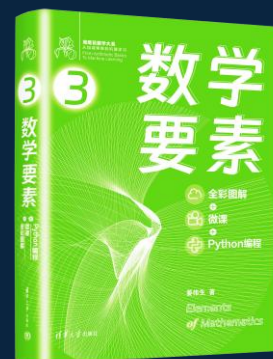
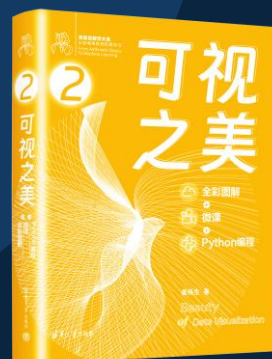
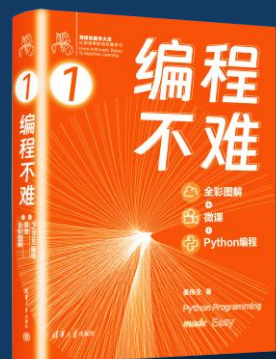
- ① Helping readers truly understand and use linear algebra is the goal of Linear Algebra Made Easy; with the help of Python programming, geometric intuition, and Manim math animation, we hope this book makes linear algebra genuinely "easy"
- ② Learning linear algebra should never be a dry numbers game, nor a painful slog of memorizing formulas — it should be a feast for the geometric eye, an expedition deep into the jungle of numbers
- ③ Truly mastering linear algebra means being able to look at data, models, phenomena, and the world itself from a higher vantage point; this book is only a starting point, and we hope it plants a seed in every reader's mind
- ④ The beauty of mathematics lies not in complexity, but in seeing the order and essence beneath the abstraction; may every reader carry that curiosity forward and turn linear algebra into an inexhaustible tool for exploring the world

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

